BITS Machine Learning Assignment 2

M.Tech · AIML · Final Submission

1. Mandatory Submission Links

GitHub Repository: https://github.com/GaneshT1133/bits-ml-assignment-2

Live Streamlit App: https://bits-ml-assignment-2-32br9urkmmonneej96cxbn.streamlit.app/

3. GitHub README Content (Part A & B)

Problem Statement

Building an end-to-end machine-learning classification pipeline and interactive web deployment using the UCI Wine Quality dataset. The project trains and compares five classifiers to predict a binary wine-quality label and provides an interactive Streamlit app for evaluation on labeled test data.

Dataset Description

The UCI Red Wine Quality dataset contains 1,599 instances. It includes 11 original physicochemical measurements and three engineered features: `total_acidity`, `bound_sulfur_dioxide`, and `sugar_to_acidity_ratio`. These form 14 input features in total. The binary target, `quality_label`, is 1 when the original quality score is at least 6 and 0 otherwise; the original quality column is dropped.

4. Metrics Comparison Table & Observations Table

All scores below were computed on the same stratified 320-instance test partition. AUC is based on predicted class probabilities.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.7438	0.8239	0.7730	0.7368	0.7545	0.4874
Decision Tree	0.7531	0.7509	0.7614	0.7836	0.7723	0.5031
KNN	0.7438	0.8144	0.7697	0.7427	0.7560	0.4867
Naive Bayes	0.7125	0.7733	0.7484	0.6959	0.7212	0.4264
Support Vector Machine	0.7406	0.8373	0.7857	0.7076	0.7446	0.4853

Observations by Model

ML Model Name	Observation about model performance
Logistic Regression	Reached 74.38% accuracy and 0.8239 AUC, with a balanced precision/recall profile (0.7730 / 0.7368). It provides a strong, interpretable linear baseline.
Decision Tree	Had the highest accuracy (75.31%), recall (78.36%), F1 (0.7723), and MCC (0.5031) in this split. Its AUC (0.7509) is the lowest, so its threshold ranking is weaker despite its best threshold-based scores.
KNN	Matched Logistic Regression's 74.38% accuracy and achieved 0.8144 AUC. Precision and recall are close (0.7697 / 0.7427), resulting in a 0.7560 F1 score.
Naive Bayes	Scored 71.25% accuracy and 0.7212 F1, the lowest on both measures in this comparison. It is the weakest overall model on this split, while providing a useful probabilistic baseline.
Support Vector Machine	Achieved the best AUC (0.8373) and precision (0.7857), indicating the strongest probability ranking and positive-prediction precision here. Its recall is 0.7076 and accuracy is 74.06%.
Overall Winner	Decision Tree is the winner when prioritizing accuracy, recall, F1, and MCC. Support Vector Machine has the best AUC and precision. The preferred model depends on the metric priority; these results reflect one fixed stratified split and do not guarantee generalization.

Winner summary: Decision Tree leads on accuracy, recall, F1, and MCC; Support Vector Machine leads on AUC and precision. Model choice depends on the metric priority.